CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 5-01-017

WASTE DISCHARGE REQUIREMENTS

FOR
STATE OF CALIFORNIA
DEPARTMENT OF CORRECTIONS
SIERRA CONSERVATION CENTER
AND
JOSEPH MARTIN
TUOLUMNE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board), finds that:

1. On 27 November 2000, the State of California, Department of Corrections, Sierra Conservation Center, and Joseph Martin (hereafter Discharger) submitted a Report of Waste Discharge to include the Tuolumne Sprayfield in its sprayfield application area for wastewater disposal. The wastewater treatment facility is owned by the State of California. The Discharger leases 179 acres of land from Joseph Martin for effluent spray disposal.

2. The Discharger owns and operates a wastewater treatment plant approximately 4.5 miles south of New Melones Lake and east of the Stanislaus River, at 5100 O'Byrnes Ferry Road, as shown on Attachment A, which is attached hereto and made part of this Order by reference. Jamestown, the nearest community, is five miles northeast of the wastewater treatment plant. The treatment plant, storage ponds, and disposal sprayfields are in Sections 27, 33, 34 and 35, T1N, R13E, MDB&M.

3. Waste Discharge Requirements (WDRs) Order No. 99-105, adopted by the Board on 28 July 1999, prescribes requirements for a discharge from the treatment plant to land. These WDRs are being updated to allow an additional sprayfield.

4. The Board adopted Cease and Desist Order No. 87-170 on 25 September 1987 for the Discharger due to overflows from the storage reservoir to Shotgun Creek during wet weather. The Discharger still does not have enough long term storage capacity to contain all the wastewater produced at the facility. The Discharger's short-term solution to resolving the lack of winter storage capacity has been to haul treated wastewater to other storage facilities during severe wet months.

5. On 15 September 2000, the Board adopted Order No. 5-00-211, NPDES No. CA 0084611, an interim NPDES permit to discharge excess wastewater in the winter to Woods Creek until a long-term storage and disposal alternative is implemented. The wastewater will be trucked from the storage ponds to Woods Creek.

6. On 1 March 2000, the Discharger presented a long term plan to resolve the lack of winter storage capacity. The plan is to construct a 12-inch pipeline to deliver treated effluent to property owned by Jack Gardella in Chinese Camp, construct a 350 acre-foot storage reservoir on that property, and dispose of the effluent on agricultural rangeland surrounding the reservoir. A time schedule, incorporated in the Provisions of this Order, requires the Discharger to implement the long-term plan by 30 August 2002.

7. The inflow to the treatment plant primarily consists of domestic waste from barracks, cellblocks, kitchens, administration spaces, and other facilities. The Discharger is operating a newly constructed tertiary treatment system consisting of two oxidation ditches, two secondary clarifiers, a tertiary filter system with six cells, a chlorination system, an aerobic digester, and a sludge drying bed with a leachate collection system. The collected leachate is reintroduced to the headworks. The treatment system is designed to handle 0.6 mgd. The plant will be able to hydraulically handle up to 1.2 mgd during peak flows.

8. Current inmate population is between 4,000 and 4,500 inmates and the average dry weather flow (ADWF) is approximately 0.55 mgd.

9. The characteristics of the influent are as follow:

Constituents	Units
Biochemical Oxygen Demand	280 mg/l
Total Suspended Solids	370 mg/l
pH	6.6 pH
Electrical Conductivity	450 μmhos/cm
Oil & Grease	57 mg/l

10. Daily sludge accumulation has increased significantly due to the use of oxidation ditch treatment system with sludge wasting. Approximately 0.5 tons of sludge are wasted daily from the aerobic digesters and is disposed of at a landfill. The Discharger is required to submit a biosolids management plan as part of this Order.

11. The treated wastewater is stored in seven lined storage reservoirs with a combined capacity of 211.8 acre-feet including two feet of freeboard. Due to the large surface area and shallow depth of the storage reservoirs, rainwater during a normal year uses up to 23 percent of the storage reservoir capacity.

12. The disposal system consists of three irrigation sprayfields (as shown on Attachment A). The lower sprayfield (11 acres) is owned by the Discharger and is west of the storage reservoirs. Approximately 179 acres of land are leased from Mr. Joseph Martin for reclamation irrigation. The main sprayfield (139 acres) is east of storage reservoirs number 7 and 8. The Tuolumne sprayfield (40 acres) is south of the main sprayfield and north of O'Byrnes Ferry Road. Cattle

graze on all the sprayfields. Shotgun Creek runs across the main sprayfield, and past the storage reservoirs and the treatment plant.

13. The Discharger reclaims its treated wastewater on land throughout the year. The removal of nutrients from the treated wastewater during winter irrigation is limited, but the extent of removal has not been documented. Plans and specifications for the installation of a monitoring system to ascertain the effect on groundwater from the nutrient loading has been submitted by the Discharger, and has been reviewed and approved by staff. Provision No. 1b requires that, by 1 April 2001, the Discharger submit a report showing that the monitoring wells have been installed.

14. Soil depth on the site is estimated to range from one to five feet before reaching bedrock. The native soils generally consist of silty clay and clayey silt with some gravel. Groundwater depth at the treatment plant location has not been determined. There is a high possibility that groundwater movement in the area is controlled by bedrock fracture, although there is no data available to indicate to what extent fracture controlled groundwater occurs at the site. The treatment plant and the irrigation sites are in the western foothills of the Sierra Nevada geomorphic province of California. The geology along the western Sierra Nevada is characterized by a belt of strongly deformed but weakly metamorphosed rocks. Local underlying rock comprises variable formations consisting mostly of metasediments (shale, muscovite schists, hornblende schists), and metavolcanics (chlorite schists, serpentinite schists, and greenstone).

15. The Discharger operates an on-site water treatment plant to serve the staff and inmates. Alum (aluminum sulfate) and chlorine are added to the water prior to filtration. The multi-media filter is backwashed daily into two clay-lined ponds, each of 50,000-gallon capacity. The discharge to the ponds averages 57,000 gallon per day. The settled backwashed water is discharged to Shotgun Creek during wet months and is used as an irrigation source during the dry months. The discharge to Shotgun Creek and sprayfields is regulated by WDRs Order No. 95-063, NPDES No. CA0082546.

16. The California Department of Health Services has established statewide reclamation criteria in Title 22, California Code of Regulations, Section 60301, et seq. (hereafter Title 22) for the use of reclaimed water, and has developed guidelines for specific uses. These requirements implement the reclamation criteria in Title 22.

17. WDRs Order No. 99-105 required a 100-foot setback/buffer zone from the reclamation area to any drainage course. The Discharger interpreted the term "drainage course" to mean a wet drainage, but not to include the same drainage during the dry season, when it does not contain surface flow. Based on this interpretation, the Discharger has been spray irrigating near and within Shotgun Creek and its tributaries which pass through the sprayfields. However, for the purposes of this Order, the term "drainage course" shall mean any drainage course that contains surface water runoff during any portion of the year, even during periods when the drainage

course does not contain surface water. Therefore, this Order provides a timeline for submittal of a report explaining how the the Discharger will come into compliance with the drainage course setback/buffer zone found in the Reclamation Specifications.

18. The Board adopted a Water Quality Control Plan, Forth Edition, for the Sacramento River Basin and the San Joaquin River Basin (hereafter Basin Plan), which contains water quality objectives for all water of the Basin. These requirements implement the Basin Plan.

19. Surface water drainage from the facility is to Tulloch Lake via Shotgun Creek and an unnamed seasonal drainage. The site lies within the New Melones Reservoir Hydrologic Unit Area No. 534.21, as depicted on the interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

20. The beneficial uses of Lake Tulloch are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

21. The beneficial uses of underlying groundwater are domestic, industrial, and agricultural supply.

22. Annual precipitation in the vicinity averages 26.46 inches. The mean evaporation rate is at 78.37 inches per year. The treatment plant, the storage reservoirs, and the irrigation fields are out of the 100-year flood plain.

23. On 20 August 1994, the California Department of Corrections certified a Notice of Exemption for this project in accordance with the California Environmental Quality Act (CEQA), (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The project as approved will not have a significant effect on water quality.

24. The Board has considered anti-degradation pursuant to State Board Resolution No. 68-16 and finds that not enough data exists to determine whether this discharge is consistent with those provisions. Therefore, this Order provides a timeline for data collection to determine whether the discharge will cause an increase in groundwater constituents above that of background levels. If the discharge is causing such an increase, then the Discharger may be required to cease the discharge, improve pond linings, implement source control, change the method of disposal, or take other action to prevent groundwater degradation.

25. Section 13267(b) of California Water Code provides that: "In conducting an investigation specified in subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of discharging, or who proposes to discharge within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of discharging, or who proposes to discharge waste outside of its region that could affect the quality of the waters of the state within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the board requires. The burden,

including costs of these reports, shall bear a reasonable relationship to the need for the reports and the benefits to be obtained from the reports."

26. This discharge is exempt from the requirements of Title 27, CCR, Section 20080, et seq. The exemption, pursuant to Section 20090 (b), is based on the following:

    a. the Board is issuing waste discharge requirements, and
    b. the discharge complies with the Basin Plan, and
    c. the wastewater does not need to be managed according to 22 CCR, Division 4.5, Chapter 11, as hazardous waste.

27. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

28. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 99-105 is rescinded and that the State of California, Department of Corrections, Sierra Conservation Center, and Joseph Martin, their agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

(Note: Other prohibitions, conditions, definitions, and methods of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.)

A. Discharge Prohibitions:

1. Discharge of waste to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially treated waste is prohibited.

3. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.

4. The discharge shall not cause the degradation of any water supply.

5. Discharge of waste classified as hazardous, as defined in Sections 2521(a) of Title 23, CCR, Section 2510, et seq., (hereafter Chapter 15, or 'designated', as defined in Section 13173 of the California Water Code, is prohibited.

6. Surfacing of wastewater outside or downgradient of the ponds is prohibited

7. Excessive irrigation with reclaimed water, or continued irrigation with reclaimed water during periods of precipitation, is prohibited.

8. Discharge of untreated or partially treated waste to groundwater is prohibited.

9. The discharge of any wastewater other than that from domestic sources (toilets, wash facilities, and kitchen/dining facilities) is prohibited.

10. Application of wastewater to areas different than those described in Finding No. 12 is prohibited.

11. The discharge of solvents, degreasers, or any other cleaners other than biodegradable soaps from vehicle washracks is prohibited.

B. Discharge Specifications:

1. The monthly average dry weather flow into the treatment plant shall not exceed 0.60 mgd.

2. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

3. As a means of discerning compliance with Discharge Specification No. 2, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.

4. The Discharger's wastewater treatment plant shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

5. The storage ponds shall not have a pH less than 6.5 or greater than 8.5.

6. The Discharger shall operate all systems and equipment to maximize treatment of wastewater and optimize the quality of the discharge.

7. The storage ponds shall be managed to prevent breeding of mosquitoes. In particular,

   a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

   b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.

c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

8. The freeboard in all ponds shall never be less than two feet as measured vertically from the water surface to the lowest point of overflow

9. The storage ponds shall have sufficient capacity to accommodate allowable wastewater flow, designed seasonal precipitation, and ancillary inflow and infiltration during the wet season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with the historical rainfall patterns.

10. By 1 October of each year thereafter, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. 8 and No. 9.

C. **Effluent Limitations:**

The discharge of effluent to the storage ponds in excess of the following limits is prohibited:

Constituent	Units	Monthly Average	Daily Maximum
BOD_5[1]	mg/l	40	80
Total Coliform	MPN/100 ml	23	230
Oil & Grease	mg/l		5
Settleable Solids	ml/l	0.5	1.0

[1] 5-day, Biochemical Oxygen Demand @ 20°C

D. **Reclamation Specifications:**

1. The following setback/buffer zones shall be provided at the reclamation areas:

Setback Definition	Spray Irrigation Setback Distance (ft.)
Disposal field to public roads	50
Disposal field to property lines	50
Disposal field to on-site irrigation well	100
Disposal field to domestic well	500
Disposal field to any food crop	100
Disposal field to drainage course	100

2. Reclaimed water used for irrigation shall meet the criteria contained in Title 22, Division 4, CCR (Section 60301 et. seq.).

3. Reclaimed wastewater conveyance lines shall be clearly marked as such. Reclaimed water controllers, valves, etc. shall be affixed with reclaimed water warning signs, and these and quick couplers and sprinkler heads shall be of a type, or secured in such a manner, that permits operation by authorized personnel only.

4. Public contact with the reclaimed water shall be precluded through such means as fences, signs, and other acceptable alternatives. To alert the public of the use of reclaimed water, signs with proper wording of sufficient size shall be placed at areas of access and around the perimeter of all areas used for effluent disposal.

5. Area irrigated with reclaimed water shall be managed to prevent breeding of mosquitoes. More specifically,

   a. Tail water must be returned and all applied irrigation water must infiltrate completely within 24-hours.

   b. Ditches not serving as wildlife habitat should be maintained free of emergent, marginal, and floating vegetation.

   c. Low-pressure and unpressurized pipelines and ditches accessible to mosquitoes shall not be used to store reclaimed water.

6. Reclaimed water for irrigation shall be managed to minimize erosion and runoff from the disposal area.

7. Direct or windblown spray shall be confined to the designated disposal area and prevented from contacting drinking water facilities.

8. The Discharger may not spray irrigate effluent during periods of precipitation and for at least 24 hours after cessation of precipitation, when ground is saturated, or when wind velocities exceed 30 mph.

E. Solids Disposal Requirements:

1. Collected screenings, grit, sludge, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer, and consistent with *Consolidated Regulations for Treatment, Storage, Processing, or Disposal of Solid Waste,* as set forth in Title 27, CCR, Division 2, Subdivision 1, Section 20005, et seq.

2. Storage, use and disposal of sewage sludge shall comply with existing Federal, State, and local laws and regulations, including permitting requirements and technical standards included in 40 CFR Part 503 and the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

3. Sludge and other solids shall be removed from ponds, clarifiers, etc. as needed to ensure optimal plant operation and adequate hydraulic capacity. Drying operations shall take place such that leachate does not impact groundwater or surface water.

4. If biosolids will be stored onsite between 15 October and 15 May of any year, then they shall be stored in a facility constructed in accordance with Class II surface impoundment or waste pile standards contained in Title 27 of the CCR. Such a facility shall be designed and maintained to prevent inundation or washout from a storm or flood with a 100 year return frequency. The Discharger shall collect any leachate or stormwater that comes in contact with the biosolids pile and return it to the wastewater treatment plant.

5. Biosolids may not be disposed of onsite except as a soil amendment used at agronomic rates. Before onsite disposal is permitted, the Discharger must submit a technical report containing all the information required for a Notice of Intent for Water Quality Order No. 2000-10-DWQ (or any subsequent document which replaces Order No. 2000-10-DWQ). The technical report must be submitted at least 60 days before the anticipated date of disposal.

6. Disposal of biosolids at a municipal solid waste landfill or at a permitted publicly owned treatment works is acceptable. The Discharger may also elect to dispose of its biosolids at a facility permitted under Order No. 2000-10-DWQ or at a similar facility permitted under individual WDRs. No matter where the biosolids are taken, the Discharger must comply with all sampling and analytical requirements of the entity that accepts the waste.

7. If the State Water Resources Control Board and the Regional Water Resources Control Board are given the authority to implement regulations contained in 40 CFR Part 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger shall comply with the standards and time schedules contained in 40 CFR Part 503 whether or not they have been incorporated into this Order.

F. Groundwater Limitation:

The discharge, in combination with other sources, shall not cause underlying groundwater to contain waste constituents in concentrations statistically greater than background water quality, except for coliform. For coliform, increases shall not cause the most probable number of total coliform organisms to exceed 2.2/100 ml over any 7-day period.

G. Surface Water Limitations:

The discharge shall not cause surface water downstream of the effluent sprayfields and ponds to contain waste constituents in concentrations statistically greater than background (i.e., upstream) surface water quality.

H. Provisions:

1. All of the following reports shall be submitted pursuant to Section 13267 of the California Water Code:

   a. By **1 March 2001**, the Discharger shall submit a sludge management report. The plan shall provide a detailed program and schedule for permanent disposal of all solid wastes generated at the wastewater treatment plant. Information provided shall include at least the items listed in Attachment B of this Order.

   b. By **1 March 2001**, the Discharger shall submit a plan which describes how compliance with the drainage course setback/buffer zone defined in Reclamation Specification No. 1 will be achieved. The plan shall also provide a schedule for the proposed reclamation system changes.

   c. To determine compliance with the Groundwater Limitation, the Discharger shall submit a Groundwater Well Installation Report by **1 May 2001**. The Monitoring Well Installation Report shall be prepared by a Registered Geologist and shall contain the information listed in Attachment C "*Items to be Included in a Monitoring Well Installation Report of Results*."

   d. By **1 January 2002**, the Discharger shall submit a Report of Waste Discharge, certified CEQA Document, and construction schedule for the project described in Finding No. 6. The project shall bring the Discharger into compliance with Discharge Specifications No. 8 - 10 (storage retention capacity for a 100-year rainfall season), without reliance on the interim NPDES permit, which currently allows seasonal discharge to Woods Creek. The RWD shall be prepared by a registered engineer and shall include a water balance that calculates the monthly net flow volume into the system, and compares that volume with capacity of the entire storage/disposal system. Net flow shall include influent, inflow/infiltration, design seasonal precipitation based on total annual precipitation using a 100-year return period (distributed monthly in accordance with historical rainfall patterns), and normal evaporation rates. The construction schedule contained within the RWD shall clearly show how construction shall be completed by **30 August 2002.**

2. The Discharger shall comply with the Monitoring and Reporting Program No. 5-01-017, which is a part of this Order, and any revisions thereto as ordered by the Executive Officer.

3. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving the disposal or reclamation areas, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

5. The Discharger shall submit to the Board on or before each compliance report due date the specified document, or if appropriate, a written report detailing compliance or noncompliance with the specific schedule date and task. If noncompliance is reported, then the Discharger shall state the reasons for noncompliance and shall provide a schedule to come into compliance.

6. The Discharger shall use the best practicable cost-effective control technique(s) currently available to comply with discharge limits specified in this order.

7. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge.

8. In the event of any change in control or ownership of land or waste discharge facilities presently owned or controlled by the Discharger, then the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to this office.

9. The Discharger shall comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

10. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

11. The Board will review this Order periodically and may revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 26 January 2001.

GARY M. CARLTON, Executive Officer

Attachments
JRM:01/26/01

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 5-01-017

FOR
STATE OF CALIFORNIA
DEPARTMENT OF CORRECTIONS
SIERRA CONSERVATION CENTER
AND
JOSEPH MARTIN
TUOLUMNE COUNTY

This Monitoring and Reporting Program (MRP) describes requirements for monitoring influent, effluent, storage reservoirs, surface water, disposal fields, groundwater, and biosolids. This MRP is issued pursuant to Water Code Section 13267.

The State of California, Department of Corrections, Sierra Conservation Center (Discharger) shall be responsible for implementation of this MRP and shall not implement any changes to the MRP unless and until a revised MRP is issued by the Executive Officer. Specific sample station locations shall be approved by Regional Board staff prior to implementation of sampling activities.

All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each grab sample shall be recorded on the sample chain of custody form.

INFLUENT MONITORING

The following shall constitute the influent monitoring program:

Constituent	Units	Type of Sample	Sampling Frequency	Reporting Frequency
Flow	mgd	Continuous	Daily	Monthly $\bar{Q} \leq 0.6$ mgd

EFFLUENT MONITORING

Effluent samples shall be collected downstream from the last connection through which wastes can be admitted into the storage reservoirs. Samples collected from the outlet structure of the treatment plant shall be considered to be adequately composited. The following shall constitute the effluent monitoring program:

Constituent	Units	Type of Sample	Sampling Frequency	Reporting Frequency
20°C BOD_5 [1]	mg/l	Grab	Monthly	Monthly 40/80
Total Suspended Solids	mg/l	Grab	Monthly	Monthly
Settleable Matter	ml/l	Grab	Monthly	Monthly 0.5/1.0
Total Dissolved Solids	mg/l	Grab	Monthly	Monthly
pH	pH units	Grab	Monthly	Monthly 6.5 - 8.5
Total Coliform	MPN/100 ml	Grab	Monthly	Monthly 23/230
Oil & Grease	mg/l	Grab	Monthly	Monthly DAILY MAX 5
Total Kjeldahl Nitrogen	mg/l	Grab	Monthly	Monthly
Nitrate Nitrogen	mg/l	Grab	Monthly	Monthly

[1] 5-days, Biochemical Oxygen Demand @ 20°C

STORAGE RESERVOIRS MONITORING

Monitoring of the storage reservoirs shall consist of the following:

Constituents	Units	Sampling Frequency	Reporting Frequency
Dissolved Oxygen	mg/l	Monthly	Monthly DO > 1.0
Freeboard	feet (tenths)	Daily	Monthly FB > 2'
Available Storage Volume	acre-feet	Monthly[1]	Monthly

[1] Shall be recorded daily from 1 October through 30 April

DISPOSAL FIELD MONITORING

Inspections of the spray disposal area, including that used for disposal of the water treatment plant backwash water, shall be conducted daily (during operation) and the results shall be included in the monthly monitoring report. Evidence of erosion, field saturation, runoff, or the presence of nuisance conditions shall be noted in the report. Reclaimed water shall also be monitored to ascertain loading rates at the disposal area. Monitoring of the disposal fields shall include the following:

Constituents	Units	Type of Sample	Sampling Frequency	Reporting Frequency
Total Flow	mgd	Continuous	Daily	Monthly
Flow to Each Field	mgd	Continuous	Daily	Monthly
Acreage Applied[1]	acres	Calculated	Daily	Monthly
Application Rate	gal/acre/day	Calculated	Daily	Monthly
Total Nitrogen	lbs/month	Calculated	Monthly	Monthly
	lbs/acre/day	Calculated	Monthly	Monthly

[1] The irrigation field shall be identified.

SURFACE WATER MONITORING

Surface water drainages which pass through or immediately below the irrigation sprayfields shall be monitored upstream and downstream of each sprayfield. The monitoring program shall consist of the following:

Constituents	Units	Frequency[1]	Frequency
Total Nitrogen	mg/l	Monthly	Monthly
Total Dissolved Solids	mg/l	Monthly	Monthly
Chlorine Residual	mg/l	Monthly	Monthly

[1] The surface water monitoring shall occur during those months in which water is present in drainages. The monitoring points shall be approved by the Board's staff.

GROUNDWATER MONITORING

Once installed, all new wells shall be added to the MRP, and the Discharger shall sample and analyze all wells. Prior to sampling, the groundwater elevations shall be measured and the wells shall be purged of at least three well volumes until measurements of pH and electrical conductivity have stabilized. All samples shall be collected using approved EPA methods. Depth to groundwater shall be measured to the nearest 0.01 feet. Water table elevations shall be calculated and used to determine the groundwater gradient and direction of flow. Groundwater samples shall be analyzed for the following:

Constituent	Units	Type of Sample	Sampling and Reporting Frequency
Groundwater Elevation	0.01 ft (AMSL)	Measurement	Monthly[1]
Total Dissolved Solids	mg/l	Grab	Monthly[1]
Nitrate as Nitrogen	mg/l	Grab	Monthly[1]
Ammonia as Nitrogen	mg/l	Grab	Monthly[1]
Total Coliform	MPN/100 ml	Grab	Monthly[1] 2.2/100 mL over 7 days

[1] Monthly groundwater monitoring shall continue for one year. After one year, a Phase I Monitoring Report, analyzing the results of the first year of groundwater monitoring and, perhaps, proposing a Phase II monitoring program shall be submitted. Based on that report, the groundwater monitoring program may be modified based on Phase I findings.

BIOSOLIDS MONITORING

The Discharger shall keep records regarding the quantity of biosolids generated by the treatment processes; any sampling and analytical data; the quantity of biosolids stored on site; and the quantity removed for disposal. The records shall also indicate that steps taken to reduce odor and other nuisance conditions. Records shall be stored onsite and available for review during inspections.

If biosolids are transported off-site for disposal, then the Discharger shall submit records identifying the hauling company, the amount of biosolids transported, and the location of disposal. If biosolids are disposed of onsite, then the Discharger shall submit the annual report information as contained in the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

All records shall be submitted as part of the Annual Monitoring Report.

REPORTING

LTR. rec'd. 11/10/04 auth. Assoc. Warden to sign SMRs.

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, sample type (e.g., effluent, pond, etc.), and reported analytical result for each sample are readily discernible. The data shall be summarized in such a manner to clearly illustrate compliance with waste discharge requirements and spatial or temporal trends, as applicable. The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported in the next scheduled monitoring report.

As required by the California Business and Professions Code Sections 6735, 7835, and 7835.1, all Groundwater Monitoring Reports shall be prepared under the direct supervision of a Registered Engineer or Geologist and signed by the registered professional.

Monthly Monitoring Reports

Monthly reports shall be submitted to the Regional Board on the **1st day of the second month following sampling** (i.e. the January Report is due by 1 March). At a minimum, the reports shall include:

1. Results of influent, effluent, storage reservoir, sprayfields, stream and groundwater monitoring. Data shall be presented in tabular format.

2. A narrative description of all preparatory, monitoring, sampling, and analytical testing activities for the groundwater and surface water monitoring. The narrative shall be sufficiently detailed to verify compliance with the WDR, this MRP, and the Standard Provisions and Reporting Requirements. Field logs shall be submitted for each well, documenting depth to groundwater; parameters measured before, during, and after purging; method of purging; calculation of casing volume; and total volume of water purged.

3. Calculation of groundwater elevations, an assessment of groundwater flow direction and gradient on the date of measurement, comparison of previous flow direction and gradient data, and discussion of seasonal trends if any.

4. Summary data tables of historical and current water table elevations and analytical results.

5. A scaled map showing relevant structures and features of the facility, the locations of monitoring wells and any other sampling stations, and groundwater elevation contours referenced to mean sea level datum.

6. A comparison of monitoring data to the discharge specifications, surface water and groundwater limitations, and an explanation of any violation of those requirements.

7. A calibration log verifying weekly calibration of field monitoring instruments (DO, pH, etc. meters) used to collect reported data.

Annual Monitoring Reports

An Annual Report shall be prepared as the December monthly monitoring report. The Annual Report will include all monitoring data required in the monthly schedule. The Annual Report shall be submitted to the Regional Board by **1 February** of each year and shall include the following:

1. If requested by staff, tabular and graphical summaries of all data collected during the year.

2. An evaluation of the performance of the wastewater treatment and disposal systems, as well as a forecast of the flows anticipated in the next year.

3. An evaluation of the groundwater quality beneath the wastewater facility.

4. Summary of information on the disposal of biosolids as described in the "Biosolids Monitoring" section, above.

5. If biosolids were disposed onsite, then the annual report information as contained in the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) or any subsequent document which replaces Order No. 2000-10-DWQ.

6. Whether the Discharger anticipates removing biosolids in the coming year, and if so, an anticipated schedule for cleaning, drying, and disposal.

7. A summary of the land management operation, including the total water application over the season; the total volume of wastewater applied; and the total nutrient loading from wastewater and chemical fertilizers.

8. A discussion of compliance and the corrective actions taken, as well as any planned or proposed actions needed to bring the discharge into full compliance with the waste discharge requirements.

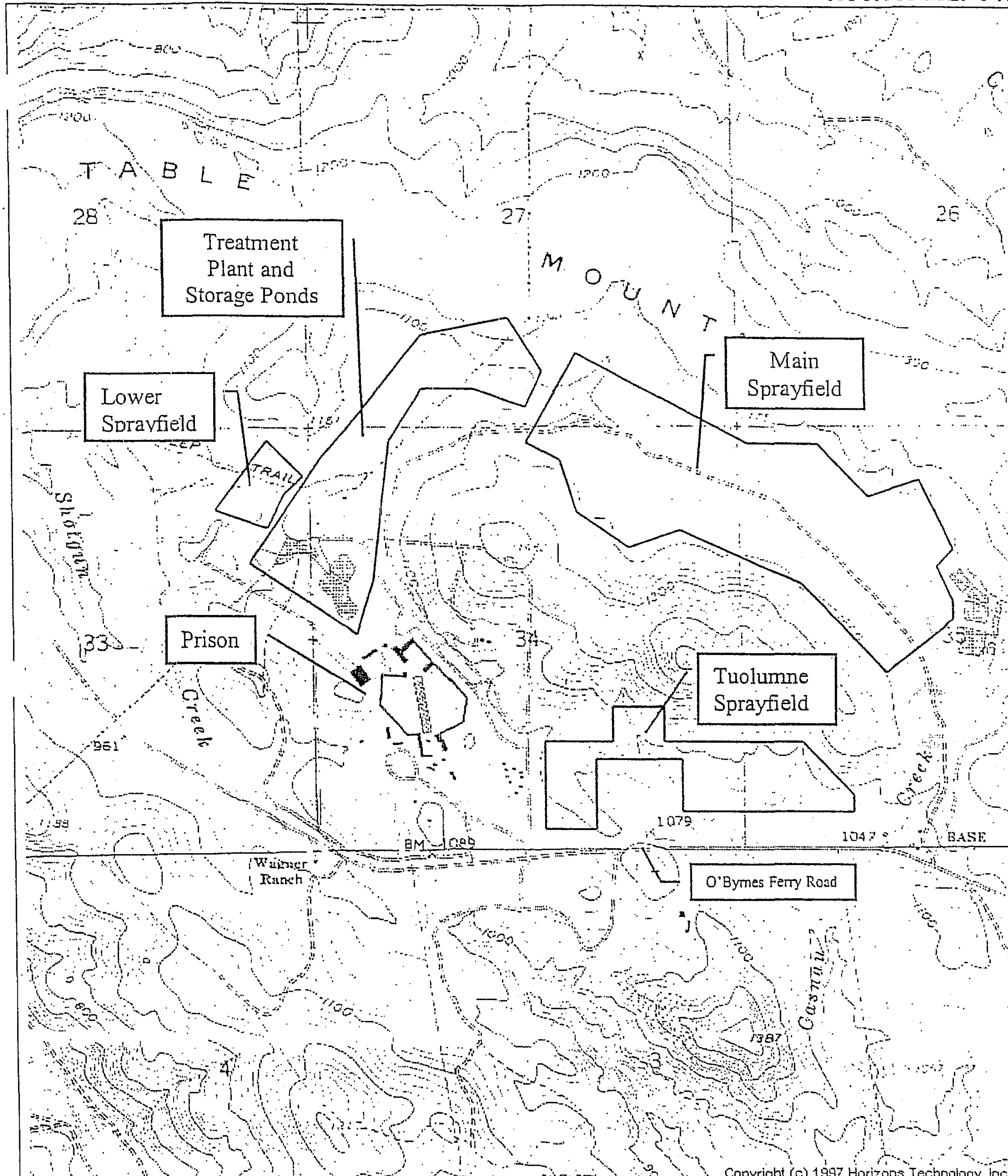
9. A discussion of any data gaps and potential deficiencies/redundancies in the monitoring system or reporting program.

The Discharger shall implement the monitoring program for influent, effluent, reclaimed wastewater, storage reservoir, disposal fields, stream monitoring and biosolids as of the date of this Order. The Discharger shall implement groundwater monitoring by 1 May 2001.

Ordered by: ______________________
GARY M. CARLTON, Executive Officer

26 January 2001
(Date)

JRM 01/26/01



Drawing Reference: U.S.G.S TOPOGRAPHIC MAP NEW MELONES DAM 7.5 MINUTE QUADRANGLE	**SITE LOCATION MAP** SIERRA CONSERVATION CENTER AND JOSEPH MARTIN TREATMENT PLANT AND SPRAYFIELDS WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-01-017	N W E S approx. scale 1 in. = 1,800 ft.

California Regional Water Quality Control Board
Central Valley Region

W n H. Hickox
Secretary for
Environmental
Protection

Sacramento Main Office
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Gray Davis
Governor

ATTACHMENT B

INFORMATION NEEDS FOR SLUDGE MANAGEMENT PLAN

A. Wastewater Treatment Facility (WWTF)

1. Provide a site map showing existing sludge handling facilities (e.g., sludge drying beds and sludge storage areas).

B. Sludge Production

1. Provide a schematic diagram showing solids flow and sludge handling operations; include, where applicable, supernatant flow and handling operations.
2. Specify annual biosolids production in dry metric tons and how this will be quantified.
3. For sludge handling facilities with sludge drying beds:
   a. Describe number and size of sludge drying beds.
   b. Describe sludge drying bed construction (e.g., liner, leachate collection system).
   c. If sludge drying beds are not lined, thoroughly describe measures taken to ensure that groundwater is not adversely affected by sludge drying operations.
   d. Indicate the frequency with which sludge is wasted and applied on sludge drying beds.

C. Biosolids Storage

1. If on-site biosolids storage is used,
   a. Describe:
      i. Size of biosolids storage area
      ii. How frequently it will be used (emergency basis only or routine use)
      iii. Typical storage duration
      iv. Leachate controls
      v. Erosion controls
      vi. Run-on/runoff controls
   b. Indicate measures that will be taken to ensure that area groundwater is not adversely affected by the biosolids storage facility
   c. For biosolids storage facilities that contain biosolids between 15 October and 15 May, describe how facilities are designed and maintained to prevent washout or inundation from a storm or flood with a return frequency of 100 years.
   d. Provide a map of showing setback distances from (where applicable)

i. Property lines

ii. Domestic water supply wells

iii. Non-domestic water supply wells

iv. Public roads and occupied onsite residences

v. Surface waters, including wetlands, creeks, ponds, lakes, underground aqueducts, and marshes

vi. Primary agricultural drainage ways

vii. Occupied non-agricultural buildings and off-site residences

viii. Domestic water supply reservoir

ix. Primary tributary to a domestic water supply

x. Domestic surface water supply intake

D. Method of Disposal

1. Describe and provide the following information related to the method of biosolids disposal.

a. Landfill Disposal

i. Name(s) and location(s) of landfill(s).

ii. Name and telephone number of the contact person at the landfill(s).

E. Grit and screenings management:

1. How much (tons or cubic yards) is generated per year.

2. How it is collected and stored prior to disposal.

3. What physical features of the plant or handling practices (if any) prevent spillage and potential contact with stormwater runoff or with surface water.

4. How it will be disposed of. Describe how this disposal method will not adversely affect water quality.

5. How frequently disposal occurs.

6. Name any contractors are involved (including transporters and disposal sites).

W n H. Hickox
ecretary for
Environmental
Protection

California Regional Water Quality Control Board

Central Valley Region

Steven T. Butler, Chair

Gray Davis
Governor

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ATTACHMENT C

ITEMS TO BE INCLUDED IN A MONITORING WELL INSTALLATION REPORT OF RESULTS

Upon installation of the monitoring wells, the Discharger shall submit a report of results, as described below. The report must be signed by a registered geologist, certified engineering geologist, or civil engineer registered or certified by the State of California.

Monitoring Well Installation Report of Results

A. Well Construction:

- Number and depth of wells drilled
- Date(s) wells drilled
- Description of drilling and construction
- Approximate locations relative to facility site(s)
- A well construction diagram for each well must be included in the report, and should contain the following details:
  - Total depth drilled
  - Depth of open hole (same as total depth drilled if no caving occurs)
  - Footage of hole collapsed
  - Length of slotted casing installed
  - Depth of bottom of casing
  - Depth to top of sand pack
  - Thickness of sand pack
  - Depth to top of bentonite seal
  - Thickness of bentonite seal
  - Thickness of concrete grout
  - Boring diameter
  - Casing diameter
  - Casing material
  - Size of perforations
  - Number of bags of sand
  - Well elevation at top of casing
  - Depth to ground water
  - Date of water level measurement
  - Monitoring well number
  - Date drilled
  - Location

B. Well Development:

- Date(s) of development of each well
- Method of development
- Volume of water purged from well
- How well development completion was determined
- Method of effluent disposal
- Field notes from well development should be included in report.

C. Well Surveying: provide reference elevations for each well and surveyor's notes

D. Water Sampling:

- Date(s) of sampling
- How well was purged
- How many well volumes purged
- Levels of temperature, EC, and pH at stabilization
- Sample collection, handling, and preservation methods
- Sample identification
- Analytical methods used
- Laboratory analytical data sheets
- Water level elevation(s)
- Groundwater contour map

E. Soil Sampling (if applicable):

- Date(s) of sampling
- Sample collection, handling, and preservation method
- Sample identification
- Analytical methods used
- Laboratory analytical data sheets

INFORMATION SHEET

WASTE DISCHARGE REQUIREMENTS ORDER NO. 5-01-017
STATE OF CALIFORNIA, DEPARTMENT OF CORRECTIONS
SIERRA CONSERVATION CENTER AND JOSEPH MARTIN
TUOLUMNE COUNTY

The State of California, Department of Corrections, Sierra Conservation Center (SCC), and Joseph Martin (hereafter Discharger) submitted a Report of Waste Discharge to revise Waste Discharge Requirements Order No. 99-105. SCC wishes to update the Order to include an additional sprayfield. SCC owns and operates a wastewater treatment plant approximately 4.5 miles south of New Melones Lake and east of Stanislaus River. Jamestown, the nearest community, is five miles northeast from the wastewater treatment plant.

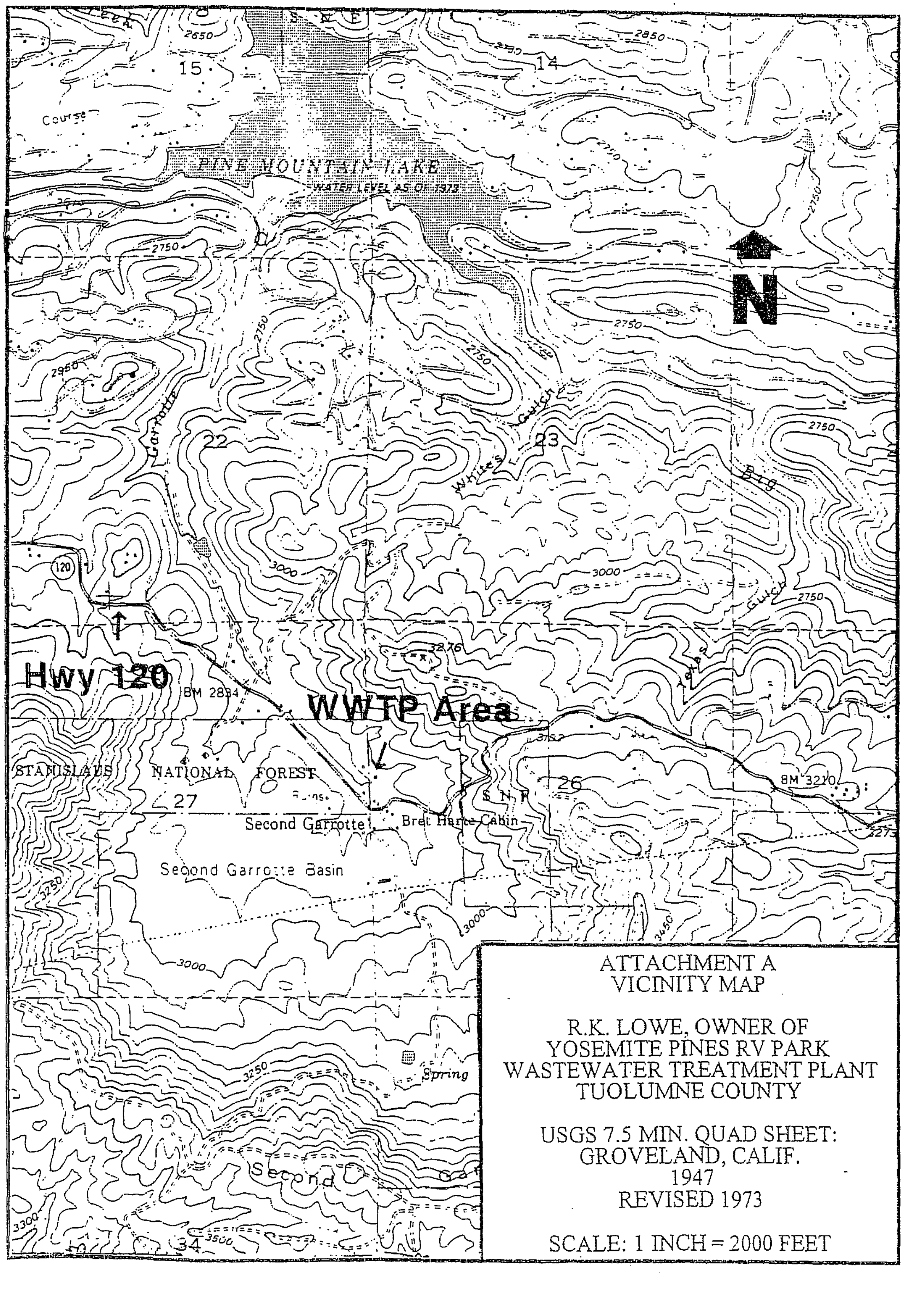
SCC is operating a newly constructed tertiary treatment system, designed to handle 0.6 million gallons per day (mgd). The plant can handle up to 1.2 mgd during peak flows. The inflow to the treatment plant primarily consists of domestic waste from barracks, cellblocks, kitchens and administration spaces. Wastewater sludge is wasted daily to drying beds, and then is hauled off-site to a landfill as needed.

The treated effluent is stored in seven lined reservoirs with a combined capacity of 212 acre-feet including two feet of freeboard. The disposal system consists of three irrigation sprayfields. The lower sprayfield (11 acres) is owned by the SCC, while the other two sprayfields (179 acres) are leased from Mr. Joseph Martin. Shotgun Creek runs across the main sprayfield and past the storage reservoirs and the treatment plant.

Current inmate population is between 4,000 and 4,500 and the average dry weather flow (ADWF) is approximately 0.55 mgd. This Order maintains an ADWF limit of 0.6 mgd. SCC does not have enough storage capacity to contain all the wastewater produced at the facility. The short-term plan to resolve the lack of winter storage capacity has been to haul treated wastewater to other storage facilities during severe wet months. On 15 September 2000, the Board adopted an interim NPDES permit to discharge, via tanker trucking, excess wastewater to Woods Creek until a long term solution to the capacity problem has been implemented. The long term plan is to construct a 12-inch pipeline to deliver treated effluent to private property in Chinese Camp, construct a 350 acre-foot storage reservoir on that property, and dispose of the effluent on agricultural rangeland surrounding the reservoir. A time schedule is incorporated in the Provisions of this Order to resolve the storage capacity problems by August 2002.

The Monitoring and Reporting Program includes requirements for monitoring of influent, effluent, the sprayfields, storage reservoirs, surface water, groundwater and biosolids. In order to determine compliance with the Groundwater Limitations, these WDRs also contain a time schedule for implementation of a groundwater monitoring program. The Order also contains a timeline for submittal of a biosolids management plan.

1/26/01:JRM



ATTACHMENT A
VICINITY MAP

R.K. LOWE, OWNER OF
YOSEMITE PINES RV PARK
WASTEWATER TREATMENT PLANT
TUOLUMNE COUNTY

USGS 7.5 MIN. QUAD SHEET:
GROVELAND, CALIF.
1947
REVISED 1973

SCALE: 1 INCH = 2000 FEET

INFORMATION SHEET

R.K. LOWE, OWNER OF
YOSEMITE PINES RV PARK
TUOLUMNE COUNTY

The Dischargers operates a 181 unit campground and Recreation Vehicle (RV) park. The facility is on a 35-acre parcel near Groveland in Tuolumne County.

Wastewater treatment and disposal is accomplished via an extended aeration package plant, with discharge to an effluent reservoir. The treatment plant has a design capacity of 22,000 gallons per day (gpd). During the summer treated wastewater is discharged to a spray field. During the winter treated wastewater is discharged to a leachfield.

The Discharger discharges approximately 13,000 gpd during April through September, and approximately 2,000 gpd during October through March. The discharge to the treatment plant consists of RV wastes and other domestic wastes. Sludges generated from the treatment process are transported to the Tuolumne Utilities District Regional Wastewater Treatment Plant. Surface water drainage is to Garrotte Creek, tributary to Pine Mountain Lake.

9 August 1996/MRB:dlk